

Model Training and Conversion

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1. Model Training
2. Model Conversion

After completing the dataset annotation tutorial content, we can use the main board to start training the model.

This tutorial only introduces CLI model training and conversion. You can refer to the official website for Python case modifications.

1. Model Training

Use CLI commands to directly train the model: Copy the yolo26n.pt file to the directory where the configuration file is located, then open a terminal in the configuration file directory:

```
cd /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/
```

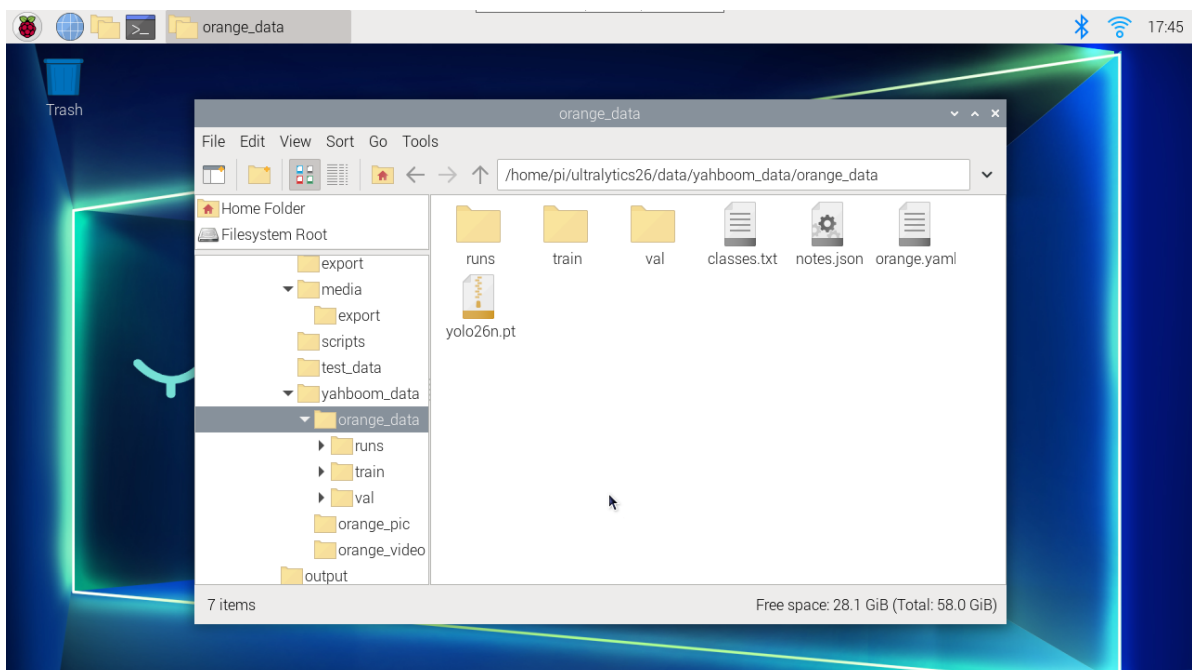
```
yolo detect train data=orange.yaml model=/home/jetson/ultralytics/yolo26n.pt  
epochs=100 imgsz=640
```

data: Dataset configuration file

model: Pre-trained model file

epochs: Number of training epochs

imgsz: Specify input image size



```

Epoch 97/100 GPU mem 2.43G box_loss 0.2911 cls_loss 0.2956 dfl_loss 0.08472 Instances 9 Size 640: 100% 10/10 1.7it/s 5.8s
Class Images 20 20 Box(P R mAP50 mAP50-95): 100% 1/1 3.1it/s 0.3s
all 20 20 0.946 1 0.993 0.933

Epoch 98/100 GPU mem 2.43G box_loss 0.2788 cls_loss 0.2739 dfl_loss 0.084257 Instances 9 Size 640: 100% 10/10 1.6it/s 6.2s
Class Images 20 20 Box(P R mAP50 mAP50-95): 100% 1/1 3.0it/s 0.3s
all 20 20 0.978 1 0.995 0.948

Epoch 99/100 GPU mem 2.43G box_loss 0.2476 cls_loss 0.294 dfl_loss 0.083827 Instances 9 Size 640: 100% 10/10 1.7it/s 5.8s
Class Images 20 20 Box(P R mAP50 mAP50-95): 100% 1/1 3.4it/s 0.3s
all 20 20 0.999 0.95 0.993 0.954

Epoch 100/100 GPU mem 2.43G box_loss 0.2482 cls_loss 0.2902 dfl_loss 0.083791 Instances 9 Size 640: 100% 10/10 1.7it/s 5.9s
Class Images 20 20 Box(P R mAP50 mAP50-95): 100% 1/1 3.3it/s 0.3s
all 20 20 0.997 1 0.995 0.953

100 epochs completed in 0.224 hours.
Optimizer stripped from /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train2/weights/last.pt, 5.4MB
Optimizer stripped from /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train2/weights/best.pt, 5.4MB

Validating /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train2/weights/best.pt...
Ultralytics 8.4.24 Python-3.10.12 torch-2.5.0a0+872d972e41.nv24.08 CUDA:0 (Orin, 7620MiB)
YOLOv8n summary (fused): 122 layers, 2,375,031 parameters, 0 gradients, 5.2 GFLOPs
Class Images Instances Box(P R mAP50 mAP50-95): 100% 1/1 2.8it/s 0.4s
all 20 20 0.997 1 0.995 0.962

Results saved to /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train2
jetson@yahboom:~/ultralytics/ultralytics/data/yahboom_data/orange_data$

```

2. Model Conversion

The final model will be generated in the runs folder: Generally select the best.pt file for use

```
/home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train2/weights
```

Convert pt model to engine model:

```
cd
/home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train2/weights
```

```
yolo export model=best.pt format=engine
```

```

TensorRT: starting export with TensorRT 10.7.0...
[03/21/2026-21:04:20] [TRT] [I] [MemUsageChange] Init CUDA: CPU -1, GPU +0, now: CPU 818, GPU 3693 (MiB)
[03/21/2026-21:04:23] [TRT] [I] [MemUsageChange] Init builder kernel library: CPU +980, GPU +632, now: CPU 1837, GPU 4368 (MiB)
[03/21/2026-21:04:23] [TRT] [I] -----
[03/21/2026-21:04:23] [TRT] [I] Input filename: best.onnx
[03/21/2026-21:04:23] [TRT] [I] ONNX IR version: 0.0.8
[03/21/2026-21:04:23] [TRT] [I] Opset version: 17
[03/21/2026-21:04:23] [TRT] [I] Producer name: pytorch
[03/21/2026-21:04:23] [TRT] [I] Producer version: 2.5.0
[03/21/2026-21:04:23] [TRT] [I] Domain:
[03/21/2026-21:04:23] [TRT] [I] Model version: 0
[03/21/2026-21:04:23] [TRT] [I] Doc string:
[03/21/2026-21:04:23] [TRT] [I] -----
TensorRT: input "images" with shape(1, 3, 640, 640) DataType.FLOAT
TensorRT: output "output0" with shape(1, 300, 6) DataType.FLOAT
TensorRT: building FP32 engine as best.engine
[03/21/2026-21:04:23] [TRT] [I] Local timing cache in use. Profiling results in this builder pass will not be stored.
[03/21/2026-21:06:02] [TRT] [I] Compiler backend is used during engine build.
[03/21/2026-21:07:15] [TRT] [I] [GraphReduction] The approximate region cut reduction algorithm is called.
[03/21/2026-21:07:15] [TRT] [I] Detected 1 inputs and 1 output network tensors.
[03/21/2026-21:07:17] [TRT] [I] Total Host Persistent Memory: 625568 bytes
[03/21/2026-21:07:17] [TRT] [I] Total Device Persistent Memory: 0 bytes
[03/21/2026-21:07:17] [TRT] [I] Max Scratch Memory: 2764800 bytes
[03/21/2026-21:07:17] [TRT] [I] [BlockAssignment] Started assigning block shifts. This will take 257 steps to complete.
[03/21/2026-21:07:17] [TRT] [I] [BlockAssignment] Algorithm ShiftTopDown took 31.6626ms to assign 11 blocks to 257 nodes requiring 18259456 bytes.
[03/21/2026-21:07:17] [TRT] [I] Total Activation Memory: 18259456 bytes
[03/21/2026-21:07:17] [TRT] [I] Total Weights Memory: 9702724 bytes
[03/21/2026-21:07:17] [TRT] [I] Compiler backend is used during engine execution.
[03/21/2026-21:07:17] [TRT] [I] Engine generation completed in 174.127 seconds.
[03/21/2026-21:07:17] [TRT] [I] [MemUsageStats] Peak memory usage of TRT CPU/GPU memory allocators: CPU 1 MiB, GPU 50 MiB
TensorRT: export success 181.1s, saved as 'best.engine' (11.3 MB)

Export complete (183.0s)
Results saved to /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train2/weights
Predict: yolo predict task=detect model=best.engine imgsz=640
Validate: yolo val task=detect model=best.engine imgsz=640 data=orange.yaml
Visualize: https://netron.app
Learn more at https://docs.ultralytics.com/modes/export
jetson@yahboom:~/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train2/weights$

```